ASSIGNMENT – 9.1

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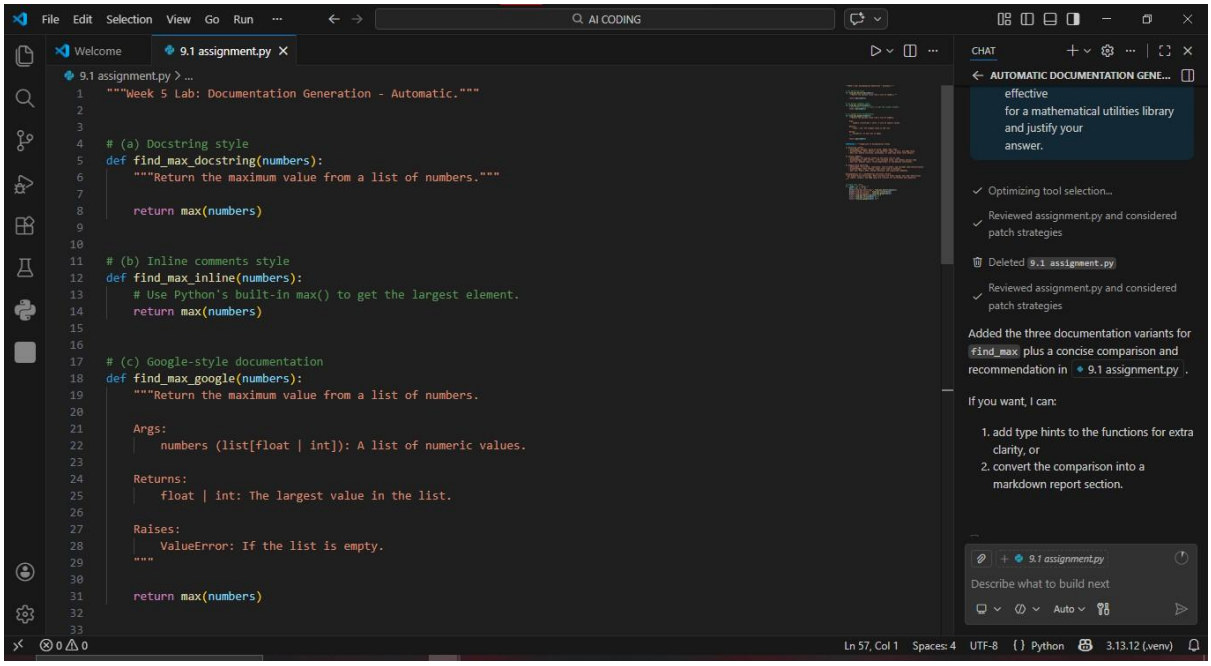
BATCH - 44

Problem 1:

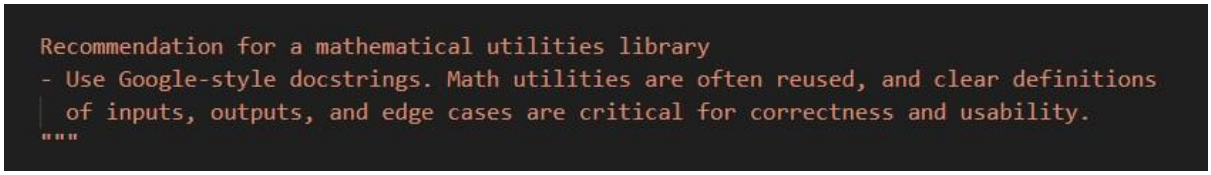
Task:

1)



2)

3)



Testcase:

```python
# Simple test cases
if __name__ == "__main__":
    data = [3, 1, 7, 2]
    print("find_max_docstring:", find_max_docstring(data))
    print("find_max_inline:", find_max_inline(data))
    print("find_max_google:", find_max_google(data))
    assert find_max_docstring(data) == 7
    assert find_max_inline(data) == 7
    assert find_max_google(data) == 7
```

Testcase output:

```
find_max_docstring: 7
find_max_inline: 7
find_max_google: 7
```

Problem 2 :

Task:

1)

```python
# Problem 2
# (a) Docstring style
def login_docstring(user, password, credentials):
    """Return True if the password matches the stored credentials."""

    return credentials.get(user) == password


# (b) Inline comments style
def login_inline(user, password, credentials):
    # Compare provided password with the stored password for the user.
    return credentials.get(user) == password


# (c) Google-style documentation
def login_google(user, password, credentials):
    """Validate a user's login attempt.

    Args:
        user (str): Username to authenticate.
        password (str): Password provided by the user.
        credentials (dict[str, str]): Mapping of usernames to passwords.

    Returns:
        bool: True when the password matches, otherwise False.
    """

    return credentials.get(user) == password
```

2)

```
COMPARISON_LOGIN = """Comparison of documentation styles for login()

1) Docstring (short)
   - Advantages: Quick overview, low maintenance overhead.
   - Disadvantages: Lacks detail on parameters, types, and behavior.
   - Best use: Small internal helpers or obvious behavior.

2) Inline comments
   - Advantages: Highlights intent right where logic occurs.
   - Disadvantages: Can repeat obvious code and add clutter.
   - Best use: Non-obvious logic, security checks, or edge cases.

3) Google-style docstring
   - Advantages: Clear contract for new readers and tooling support.
   - Disadvantages: More verbose and slower to write.
   - Best use: Public-facing functions and onboarding materials.
```

3)

```
Recommendation for onboarding new developers
- Use Google-style docstrings. They provide consistent, structured explanations
  of inputs and outputs, which helps new contributors understand expectations
  and reduces misinterpretation.
"""
```

Testcase:

```
# testcases for login functions
creds = {"alice": "p@ss", "bob": "1234"}
print("login_docstring:", login_docstring("alice", "p@ss", creds))
print("login_inline:", login_inline("bob", "wrong", creds))
print("login_google:", login_google("bob", "1234", creds))
assert login_docstring("alice", "p@ss", creds) is True
assert login_inline("bob", "wrong", creds) is False
assert login_google("bob", "1234", creds) is True
```

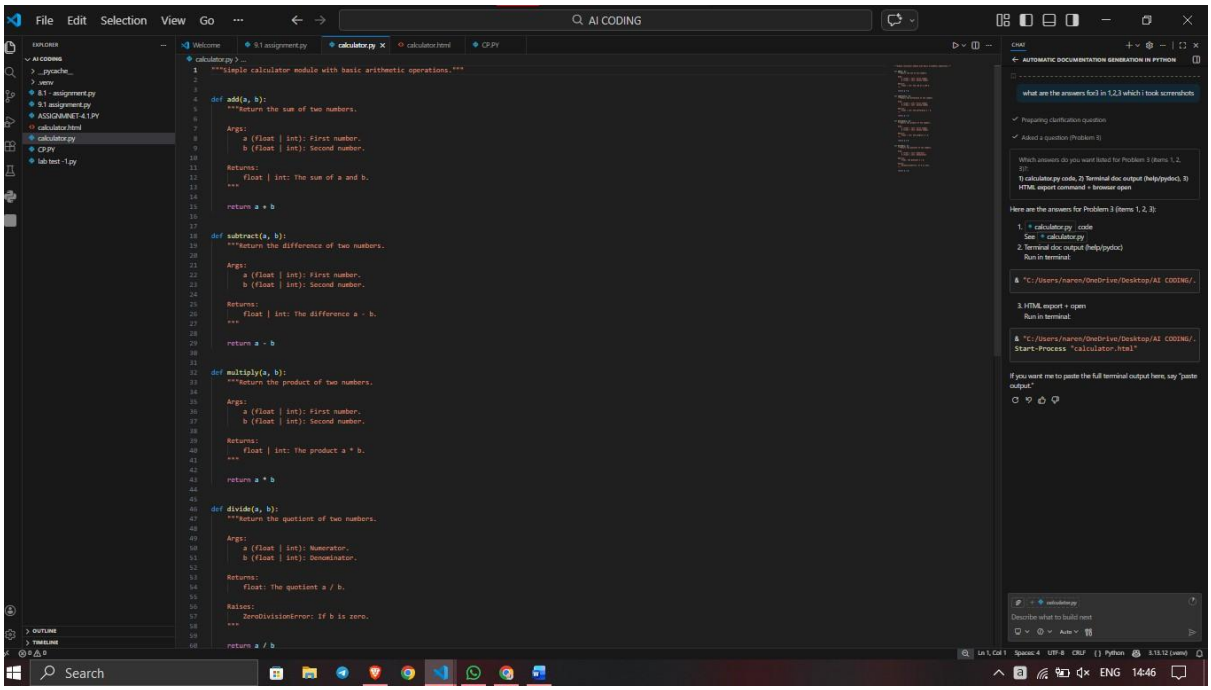
Testcase Output:

```
login_docstring: True
login_inline: False
login_google: True
```

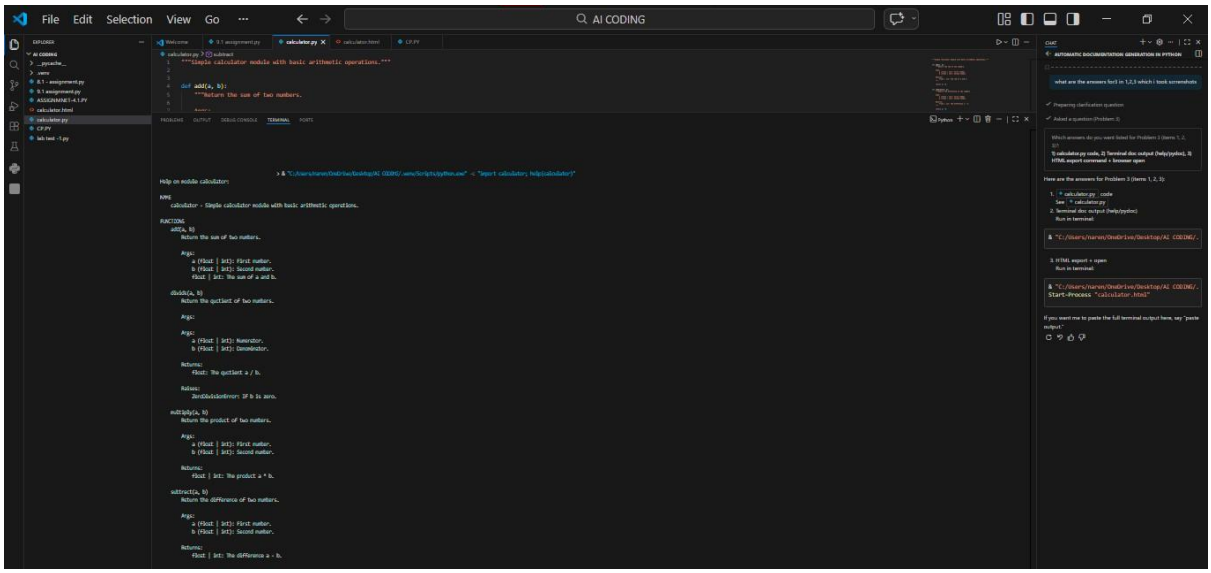
Problem 3 :

Task:

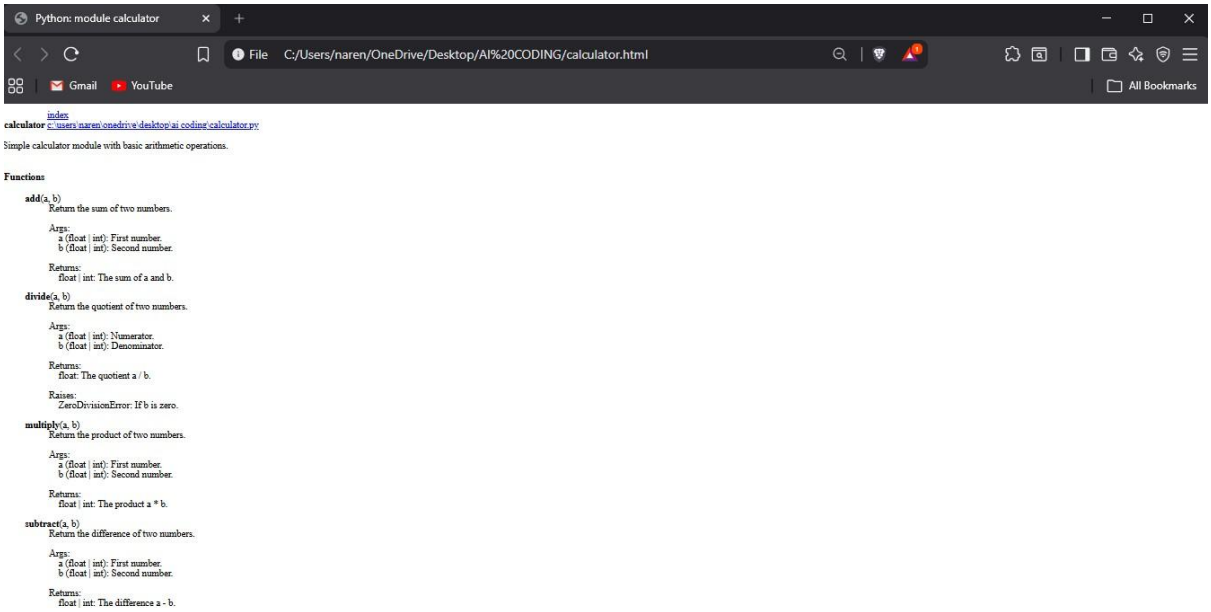
1)


File Edit Selection View Go
AI CODING
EXPLORER
AI CODING
__pycache__
.venv
8.1 - assignment.py
9.1 assignment.py
ASSIGNMNET-4.1.PY
calculator.html
calculator.py
CP.PY
lab test -1.py
Welcome
9.1 assignment.py
calculator.py
calculator.html
CP.PY
calculator.py > ...
"""Simple calculator module with basic arithmetic operations."""
def add(a, b):
"""Return the sum of two numbers.
Args:
a (float | int): First number.
b (float | int): Second number.
Returns:
float | int: The sum of a and b.
"""
return a + b
def subtract(a, b):
"""Return the difference of two numbers.
Args:
a (float | int): First number.
b (float | int): Second number.
Returns:
float | int: The difference a - b.
"""
return a - b
def multiply(a, b):
"""Return the product of two numbers.
Args:
a (float | int): First number.
b (float | int): Second number.
Returns:
float | int: The product a * b.
"""
return a * b
def divide(a, b):
"""Return the quotient of two numbers.
Args:
a (float | int): Numerator.
b (float | int): Denominator.
Returns:
float: The quotient a / b.
Raises:
ZeroDivisionError: If b is zero.
"""
return a / b
CHAT
AUTOMATIC DOCUMENTATION GENERATION IN PYTHON
what are the answers for3 in 1,2,3 which i took screenshots
Preparing clarification question
Asked a question (Problem 3)
Which answers do you want listed for Problem 3 (items 1, 2, 3)?
1) calculator.py code, 2) Terminal doc output (help/pydoc), 3) HTML export command + browser open
Here are the answers for Problem 3 (items 1, 2, 3):
1. calculator.py code
See: calculator.py
2. Terminal doc output (help/pydoc)
Run in terminal:
& "C:/Users/naren/OneDrive/Desktop/AI CODING/.
3. HTML export + open
Run in terminal:
& "C:/Users/naren/OneDrive/Desktop/AI CODING/.
Start-Process "calculator.html"
If you want me to paste the full terminal output here, say "paste output."
Describe what to build next
Search
ENG 14:46

2)


File Edit Selection View Go
AI CODING
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & "C:/Users/naren/OneDrive/Desktop/AI CODING/.venv/Scripts/python.exe" -c "import calculator; help(calculator)"
Help on module calculator:
NAME
calculator - Simple calculator module with basic arithmetic operations.
FUNCTIONS
add(a, b)
Return the sum of two numbers.
Args:
a (float | int): First number.
b (float | int): Second number.
Returns:
float | int: The sum of a and b.
divide(a, b)
Return the quotient of two numbers.
Raises:
ZeroDivisionError: If b is zero.
multiply(a, b)
Return the product of two numbers.
subtract(a, b)
Return the difference of two numbers.

3)



Problem 4:

1)

```python
def decimal_to_binary(n):


    if n < 0:
        raise ValueError("n must be non-negative")
    return bin(n)[2:]


def binary_to_decimal(b):


    if not b or any(ch not in "01" for ch in b):
        raise ValueError("b must be a non-empty binary string")
    return int(b, 2)


def decimal_to_hexadecimal(n):


    if n < 0:
        raise ValueError("n must be non-negative")
    return hex(n)[2:]
```

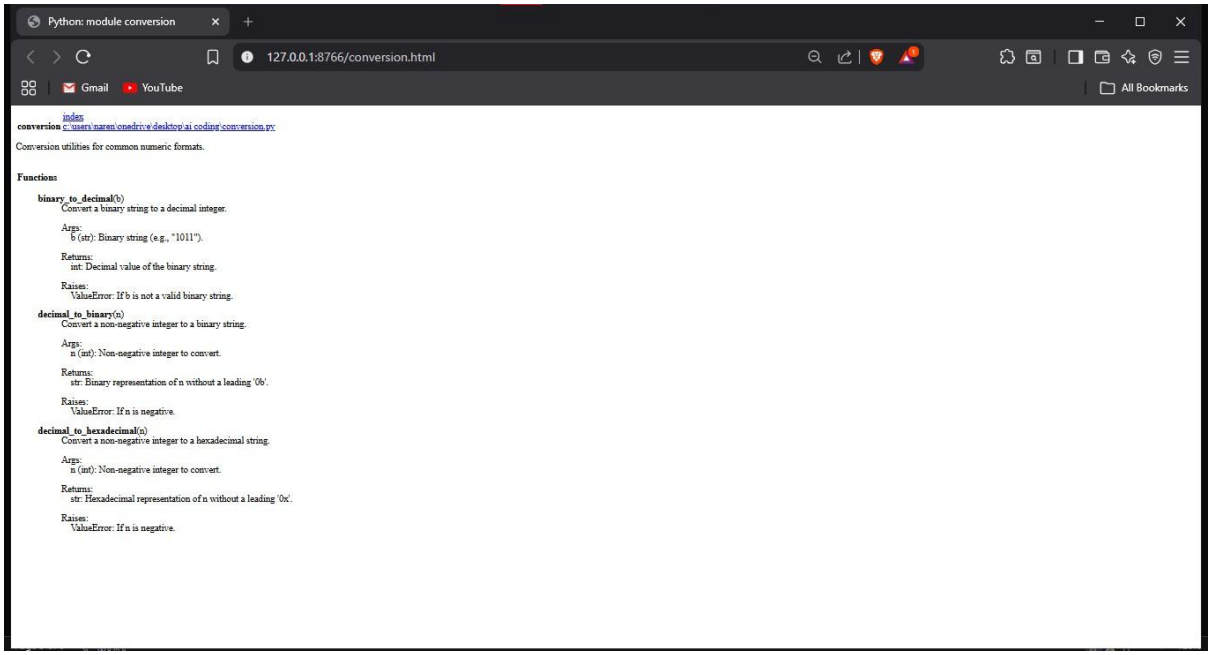
2)

```python
"""Conversion utilities for common numeric formats."""


def decimal_to_binary(n):
    """Convert a non-negative integer to a binary string.

    Args:
        n (int): Non-negative integer to convert.

    Returns:
        str: Binary representation of n without a leading '0b'.

    Raises:
        ValueError: If n is negative.
    """

    if n < 0:
        raise ValueError("n must be non-negative")
    return bin(n)[2:]


def binary_to_decimal(b):
    """Convert a binary string to a decimal integer.

    Args:
        b (str): Binary string (e.g., "1011").

    Returns:
        int: Decimal value of the binary string.

    Raises:
        ValueError: If b is not a valid binary string.
    """

    if not b or any(ch not in "01" for ch in b):
        raise ValueError("b must be a non-empty binary string")
    return int(b, 2)


def decimal_to_hexadecimal(n):
    """Convert a non-negative integer to a hexadecimal string.

    Args:
        n (int): Non-negative integer to convert.

    Returns:
        str: Hexadecimal representation of n without a leading '0x'.

    Raises:
        ValueError: If n is negative.
    """

    if n < 0:
        raise ValueError("n must be non-negative")
    return hex(n)[2:]
```

3)

```
Python Library Documentation: module conversion

NAME
    conversion - Conversion utilities for common numeric formats.

FUNCTIONS
    binary_to_decimal(b)
        Convert a binary string to a decimal integer.

        Args:
            b (str): Binary string (e.g., "1011").

        Returns:
            int: Decimal value of the binary string.

        Raises:
            ValueError: If b is not a valid binary string.

    decimal_to_binary(n)
        Convert a non-negative integer to a binary string.

        Args:
            n (int): Non-negative integer to convert.

        Returns:
            str: Binary representation of n without a leading '0b'.

        Raises:
            ValueError: If n is negative.

    decimal_to_hexadecimal(n)
        Convert a non-negative integer to a hexadecimal string.

        Args:
            n (int): Non-negative integer to convert.

        Returns:
            str: Hexadecimal representation of n without a leading '0x'.

        Raises:
            ValueError: If n is negative.
```

4)



Problem 5:

Task:

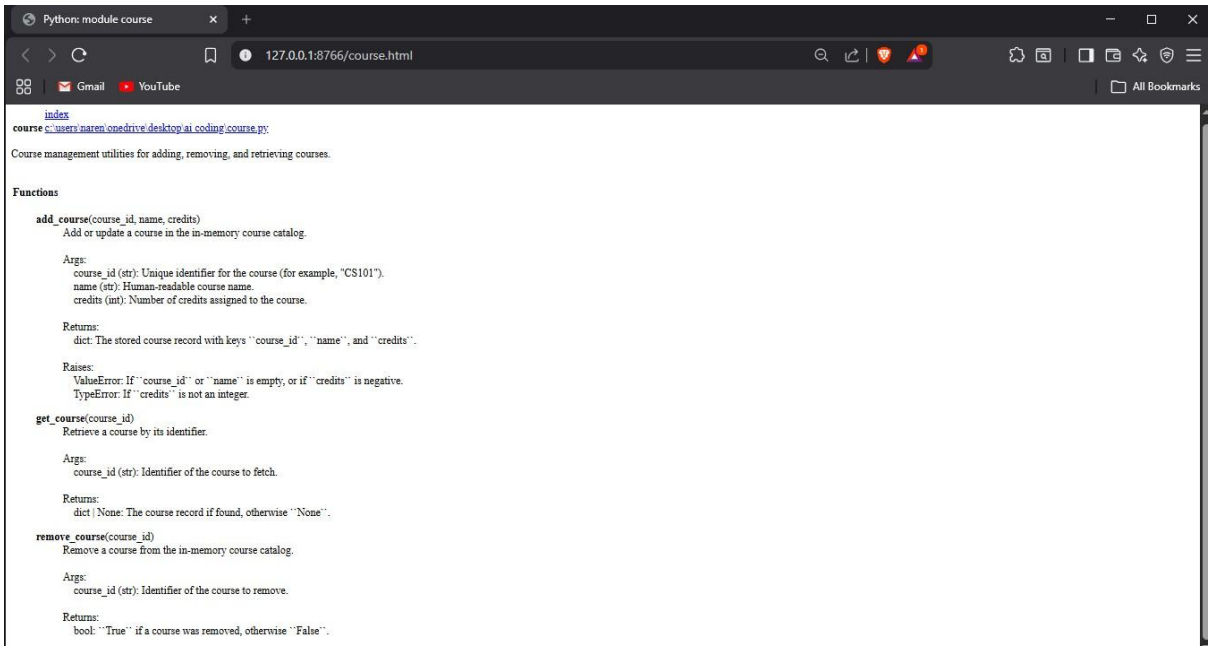
1)

```python
_courses = {}


def add_course(course_id, name, credits):


    if not course_id or not str(course_id).strip():
        raise ValueError("course_id must be a non-empty value")
    if not name or not str(name).strip():
        raise ValueError("name must be a non-empty value")
    if not isinstance(credits, int):
        raise TypeError("credits must be an integer")
    if credits < 0:
        raise ValueError("credits must be non-negative")

    record = {
        "course_id": str(course_id).strip(),
        "name": str(name).strip(),
        "credits": credits,
    }
    _courses[record["course_id"]] = record
    return record


def remove_course(course_id):


    if not course_id:
        return False
    return _courses.pop(str(course_id).strip(), None) is not None


def get_course(course_id):


    if not course_id:
        return None
    return _courses.get(str(course_id).strip())
```

2)

```python
"""Course management utilities for adding, removing, and retrieving courses."""

_courses = {}


def add_course(course_id, name, credits):
    """Add or update a course in the in-memory course catalog.

    Args:
        course_id (str): Unique identifier for the course (for example, "CS101").
        name (str): Human-readable course name.
        credits (int): Number of credits assigned to the course.

    Returns:
        dict: The stored course record with keys ``course_id``, ``name``, and ``credits``.

    Raises:
        ValueError: If ``course_id`` or ``name`` is empty, or if ``credits`` is negative.
        TypeError: If ``credits`` is not an integer.
    """

    if not course_id or not str(course_id).strip():
        raise ValueError("course_id must be a non-empty value")
    if not name or not str(name).strip():
        raise ValueError("name must be a non-empty value")
    if not isinstance(credits, int):
        raise TypeError("credits must be an integer")
    if credits < 0:
        raise ValueError("credits must be non-negative")

    record = {
        "course_id": str(course_id).strip(),
        "name": str(name).strip(),
        "credits": credits,
    }
    _courses[record["course_id"]] = record
    return record


def remove_course(course_id):
    """Remove a course from the in-memory course catalog.

    Args:
        course_id (str): Identifier of the course to remove.

    Returns:
        bool: ``True`` if a course was removed, otherwise ``False``.
    """

    if not course_id:
        return False
    return _courses.pop(str(course_id).strip(), None) is not None


def get_course(course_id):
    """Retrieve a course by its identifier.

    Args:
        course_id (str): Identifier of the course to fetch.

    Returns:
        dict | None: The course record if found, otherwise ``None``.
    """

    if not course_id:
        return None
    return _courses.get(str(course_id).strip())
```

3)

```
NAME
    conversion - Conversion utilities for common numeric formats.

FUNCTIONS
    binary_to_decimal(b)
        Convert a binary string to a decimal integer.

        Args:
            b (str): Binary string (e.g., "1011").

        Returns:
            int: Decimal value of the binary string.

        Raises:
            ValueError: If b is not a valid binary string.

    decimal_to_binary(n)
        Convert a non-negative integer to a binary string.

        Args:
            n (int): Non-negative integer to convert.

        Returns:
            str: Binary representation of n without a leading '0b'.

        Raises:
            ValueError: If n is negative.

    decimal_to_hexadecimal(n)
        Convert a non-negative integer to a hexadecimal string.

        Args:
            n (int): Non-negative integer to convert.

        Returns:
            str: Hexadecimal representation of n without a leading '0x'.

        Raises:
            ValueError: If n is negative.
```

4)